Schooling and college adjustments to productivity

Controls: Modern Adjustment for Human Capital; Apply College Share Adjustment; College Share Coefficient.

1. What is adjusted. For a county or MSA g, write log GDP per employment as $y_g = \log(Q_g/N_g)$, where Q_g is GDP and N_g is employment. Let S_g denote average years of schooling and C_g the college share, measured between zero and one. The ordinary human-capital adjustment changes the outcome to

$$y_g^{\text{adj}} = y_g - \beta_g(S_g - \bar{S}) - \kappa C_g.$$

Here β_g is a supplied return to schooling, $\bar{S}$ is average schooling in that model's final sample, and κ is the selected college coefficient. An inactive adjustment is omitted. The employment or density variable on the right-hand side remains unchanged.

2. The schooling choices. None leaves the schooling contribution in the outcome. Same Return uses one common return to an additional year of schooling, estimated from micro data. Specific Return uses supplied returns that vary with the place's reference-year employment density; the data build allows a quadratic relationship with log density. These returns are fixed across the modern years shown in the app. Run Both displays separate estimates for the two adjustments; it does not add them together.

The reference $\bar{S}$ is recalculated for each final estimation sample and reported with the results. With a common return, centering changes only the intercept. With a place-specific return, the coefficient is interpreted at that reference schooling level. The schooling inputs use 2009 data for MSAs and ACS 2010–2014 data for counties.

3. The college coefficient. The default is $\kappa = 0.75$, a chosen adjustment parameter. For example, a one-percentage-point higher college share subtracts $0.75 \times 0.01 = 0.0075$ from log GDP per employment. Changing the coefficient changes how much of the productivity difference is attributed to college share. The app does not estimate this coefficient in the displayed regression. Standard errors treat the supplied schooling returns and college coefficient as fixed.

4. Interpretation and the CH extension. The adjusted regression asks how density relates to productivity after removing the specified human-capital contributions. It does not separately identify their causal effects. These outcome adjustments are available for ordinary County and MSA models and the MSA geography-only CH option. In Geography + human capital CH, schooling and college instead enter the county contributions to the index; the aggregate outcome then remains unadjusted to avoid counting the same correction twice.